

Power-System Evolution and Symbolic Representation

From the changing electric grid to a compact engineering model

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Two views of the same power system

LECTURE STRUCTURE

Evolution and state of the art

- how local grids became interconnected networks;
- why sensing, communication, storage, and distributed resources matter;
- how generation mix and resource geography shape operation.

Symbolic representation

- how physical equipment becomes a one-line diagram;
- what each symbol preserves and suppresses;
- how buses, branches, injections, and withdrawals form a solvable model.

The grid keeps changing, but engineers still need a stable language for representing connectivity and power flow.

PART I

Power-System Evolution and State of the Art

How the grid grew, where electricity comes from, and why operation is changing

Acknowledgment

This section is adapted primarily from Prof. Anil Pahwa's
Electricity Access Challenge 2025 slides.

From local service to interconnection

POWER-SYSTEM EVOLUTION



1882: local DC service

Pearl Street served a compact area in lower Manhattan.



1884: early AC transmission

Transformers made higher-voltage transfer practical.



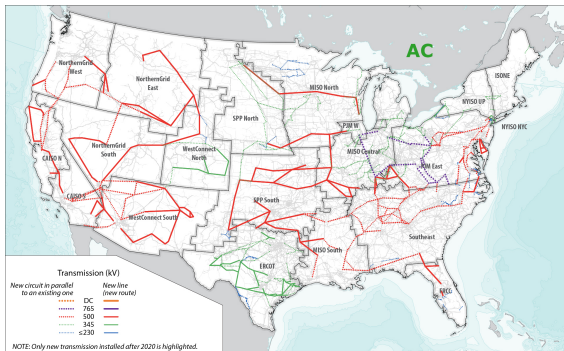
Today: interconnected infrastructure

Power systems coordinate many sources, networks, and users.

Increasing distance, scale, and interconnection created the need for common equipment models and coordinated operation.

National grid

CONTINENTAL-SCALE COORDINATION



NREL planning-study map of regional interfaces and candidate AC transmission expansion.

Bulk transmission links geographically separated resources and demand while requiring planning across regional boundaries.

Network reliability

STATE OF THE GRID



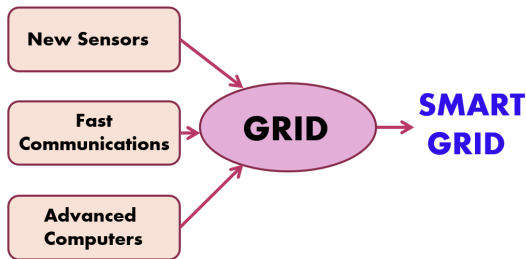
Why local failures can spread

- power flows redistribute immediately after an outage;
- overloaded equipment and inadequate visibility can create cascading events;
- aging assets, extreme weather, cyber risk, and changing generation add new operating stress.

Reliability depends on the condition and coordination of the whole interconnected network.

Smart-grid layers

SMART-GRID CONCEPT



Core capabilities

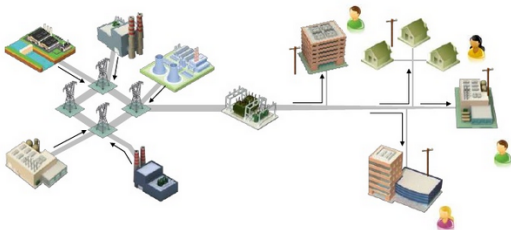
- sensors make operating conditions visible;
- communication moves measurements and commands;
- computation supports estimation, protection, optimization, and control.

“Smart” describes coordinated sensing, communication, and decision-making—not a different physical law.

Traditional and evolving grids

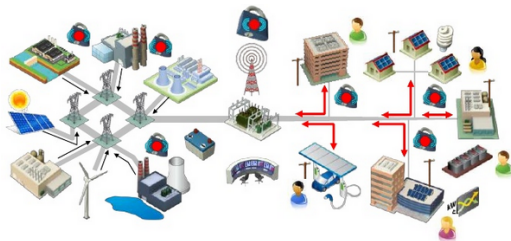
TRADITIONAL STRUCTURE AND EMERGING OPERATION

Traditional grid



Large central plants; mostly one-way delivery; limited end-use visibility.

Evolving grid



Distributed energy, storage, communication, and more active customers.

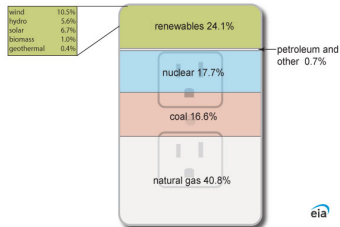
The same network must increasingly coordinate power that enters, exits, and changes direction at many locations.

Generation mix

UNITED STATES AND KANSAS

Sources of U.S. electricity generation, 2025

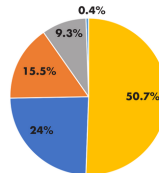
total=4.43 trillion kilowatthours



Data source: U.S. Energy Information Administration, *Electric Power Monthly*, February 2026, preliminary data

Note: Includes electricity generation from power plants with at least 1,000 kilowatts of electric generation capacity (utility-scale). Hydro is conventional hydroelectric. Petroleum includes petroleum liquids, petroleum coke, other gases (blast furnace gas and other manufactured and waste gases derived from fossil fuels), hydroelectric pumped storage, and other sources (non-biogenic municipal solid waste, batteries, hydrogen, purchased steam, sulfur, tire-derived fuel, and other miscellaneous energy sources). Sum of percentages may not equal 100% because of independent rounding.

Kansas Electric Generation Sources by Total Megawatt hours



- Wind - 30,142,619 MWh
- Coal - 14,253,477 MWh
- Nuclear - 9,222,831 MWh
- Natural Gas - 5,538,953 MWh
- Other - 257,424 MWh

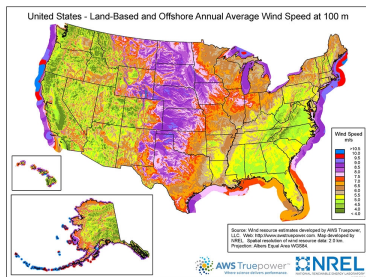
Source: U.S. Department of Energy, The Energy Information Administration (EIA) August 2023 to July 2024

A national portfolio can be diverse even when an individual state relies heavily on a particular local resource.

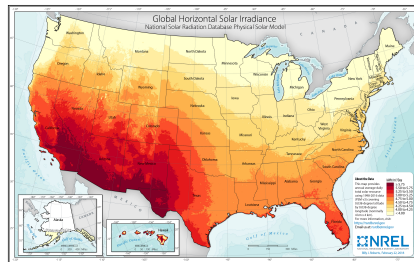
Resource geography

RESOURCE GEOGRAPHY

Wind resource



Solar resource

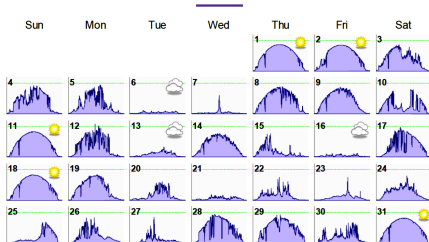


Resource maps help explain why new generation and new transmission do not appear uniformly across the country.

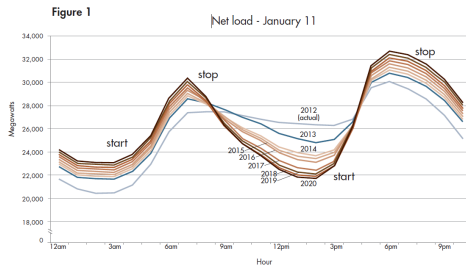
Variable generation and net load

FROM AVAILABLE ENERGY TO NET-LOAD RAMPS

PV output varies



The system sees net load



Operators must balance demand minus variable generation continuously, including steep ramps and forecast error.

PART II

Symbolic Representation

From physical infrastructure to a compact one-line model

What must a drawing preserve so that engineers can analyze the system?

Starting problem

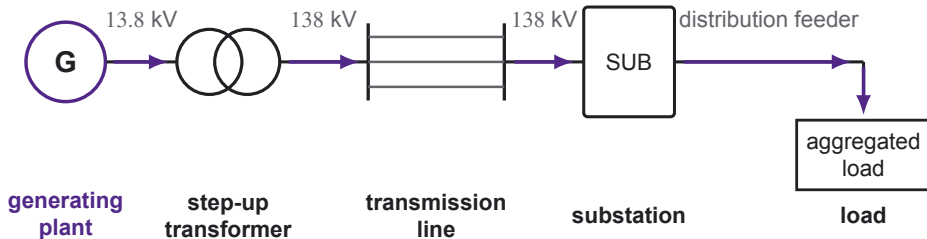
STARTING PROBLEM

A generator supplies a campus through transformers and a transmission line. What information must a compact drawing preserve before an engineer can study the system?



Physical path

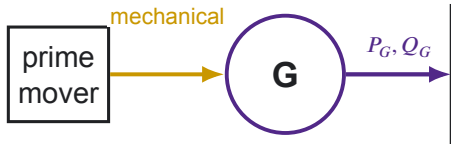
THE PHYSICAL PATH



Interconnection lets power travel; voltage transformation makes that transfer practical.

Generator

OBJECT CLASS 1 — GENERATOR



physical machine → generator symbol

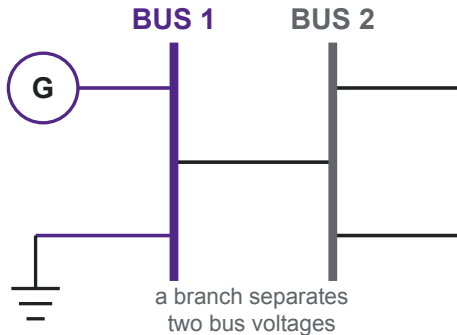
What the object preserves

- terminal bus and nominal voltage;
- real/reactive capability and control target;
- in-service status and equipment rating.

The circle labeled G represents an energy-conversion device, not an ideal source with unlimited output.

Bus

OBJECT CLASS 2 — BUS



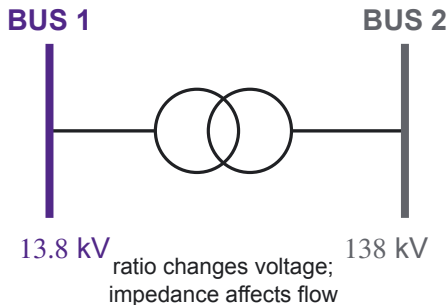
Bus meaning

- Every terminal on one bus shares the same voltage phasor.
- A branch between buses permits different voltage magnitudes or angles.
- The bus organizes connectivity; it is not a storage tank for power.

One bus ID corresponds to one electrical node and one solved voltage.

Transformer

OBJECT CLASS 3 — TRANSFORMER



Essential records

- bus endpoints and voltage ratio;
- impedance, rating, and tap position;
- power flow, loss, and loading results.

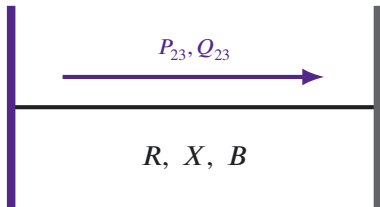
A transformer transfers AC power while changing the voltage-current scale.

Transmission line

OBJECT CLASS 4 — LINE

BUS 2

BUS 3



length + conductor geometry $\rightarrow$ electrical parameters

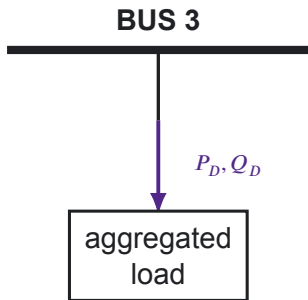
Essential records

- from bus, to bus, and circuit ID;
- resistance, reactance, charging, and rating;
- end flows, loss, current, and loading.

**The line is a two-ended branch;
direction labels make its reported
flow unambiguous.**

Load

OBJECT CLASS 5 — LOAD



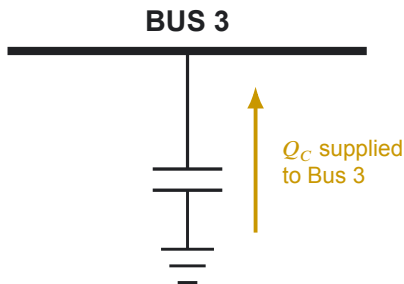
What aggregation means

- Many motors, lights, electronics, and buildings become one demand record.
- Positive P_D and Q_D denote consumption in this course.
- Demand changes with operating condition and time.

A load arrow is a declared consumption reference, not a statement that power can never reverse.

Shunt

OBJECT CLASS 6 — SHUNT



Capacitor or reactor

- A capacitor supplies reactive power at its bus.
- A reactor absorbs reactive power at its bus.
- Rating, status, and bus voltage determine the operating effect.

Local shunt support can reduce the reactive power carried by an upstream branch.

Six symbols

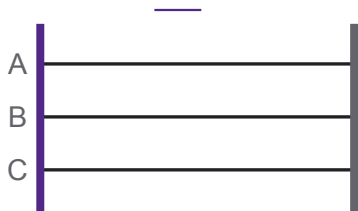
SYMBOL REFERENCE

Symbol	Object	Physical role	Typical system record
	Generator	converts primary energy	bus, rating, controls, limits
	Bus	joins terminals at one voltage	ID, nominal kV, attached objects
	Transformer	changes voltage-current scale	buses, ratio, impedance, rating
	Line	transfers power between buses	endpoints, circuit ID, $R/X/B$
	Load Shunt	consumes real/reactive power supplies or absorbs local Q	bus, P_D , Q_D , status bus, Mvar rating, status

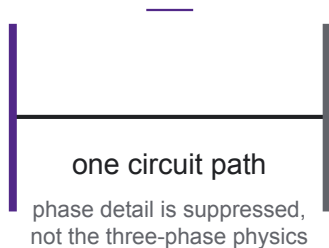
Three-phase system, one drawn path

THREE-PHASE SYSTEM, ONE DRAWN PATH

Physical three-phase conductors



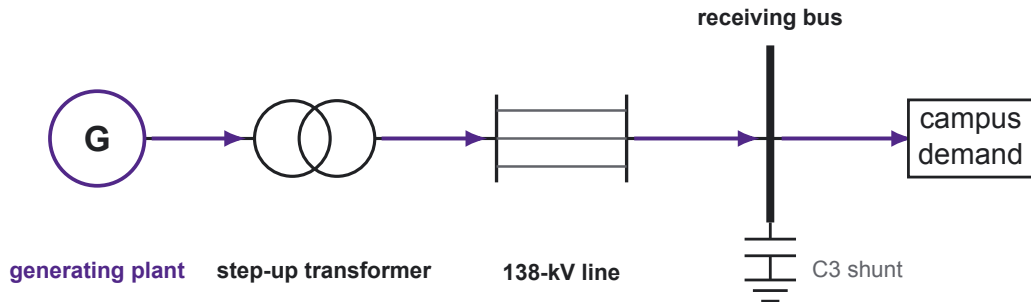
Balanced-system one-line



A one-line is a topology-and-model drawing, not a literal picture of every conductor.

Plant to campus

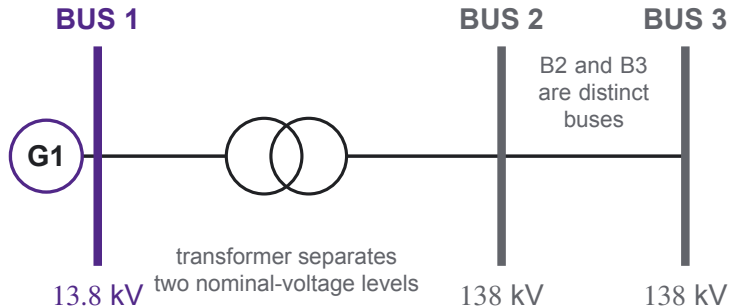
PLANT TO CAMPUS



The physical story already contains all six object classes; the next steps make their electrical relationships explicit.

Define voltage zones

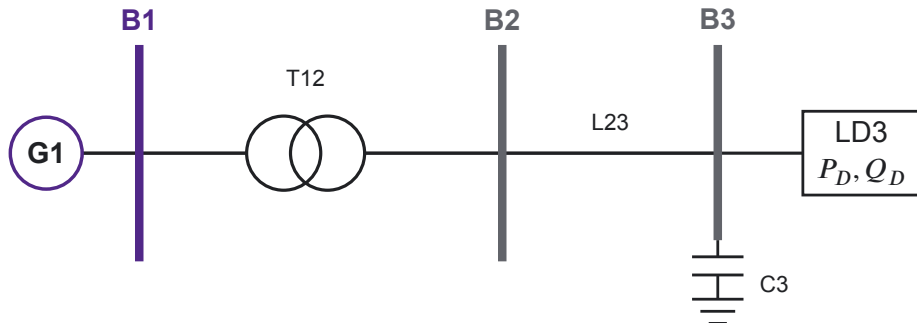
STEP 1: PLACE BUSES AT DISTINCT ELECTRICAL NODES



A new bus is required whenever the network needs a distinct voltage state.

Preserve endpoints and roles

STEP 2: REPLACE EACH DEVICE WITH ITS ONE-LINE SYMBOL

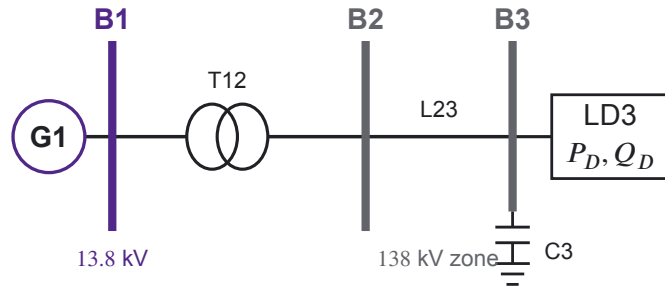


Stable object IDs make the drawing, data records, and later results refer to the same equipment.

Complete the one-line

STEP 3: VERIFY IDS, CONNECTIVITY, AND VOLTAGE ZONES

Object count



- 1 generator;
- 3 buses;
- 1 transformer;
- 1 line;
- 1 load;
- 1 shunt.

The one-line preserves identity, connectivity, and voltage zones while suppressing construction detail.

Given data and sign convention

THE POWER TRACE STARTS FROM FOUR DECLARED QUANTITIES

Receiving-end demand and support

$$S_D = (90 + j30) \text{ MVA}$$

$$Q_C = 12 \text{ Mvar} \quad \text{capacitive supply}$$

Previously determined branch absorption

$$S_{L,\text{loss}} = (2 + j8) \text{ MVA}$$

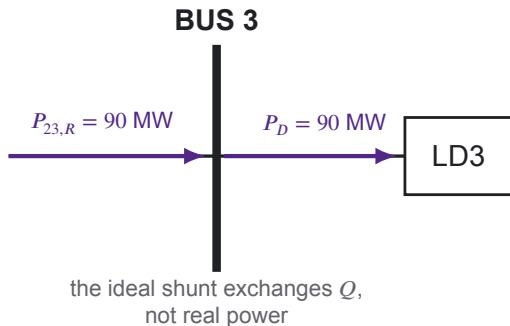
$$S_{T,\text{loss}} = (1 + j4) \text{ MVA}$$

Positive load P_D, Q_D denote consumption; positive Q_C denotes reactive power supplied by the capacitor.

Notation: j is upright; MVA applies to the complete complex-power value, while MW and Mvar identify its real and reactive components.

Real-power balance at Bus 3

TRACE REAL POWER AT THE RECEIVING BUS



$$P_{23,R} + P_C = P_D$$

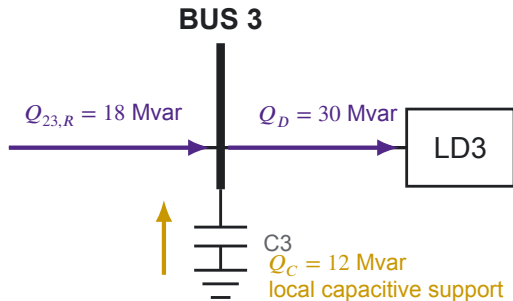
$$P_C \approx 0$$

$$P_{23,R} = 90 \text{ MW}$$

At a bus, incoming and outgoing real power must reconcile under one declared sign convention.

Reactive power at Bus 3

TRACE REACTIVE POWER AT THE RECEIVING BUS



$$Q_{23,R} + Q_C = Q_D$$

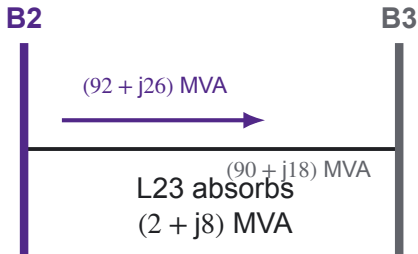
$$Q_{23,R} + 12 = 30$$

$$Q_{23,R} = 18 \text{ Mvar}$$

The load still absorbs 30 Mvar, but the transmission line delivers only the 18-Mvar remainder.

Line sending-end power

MOVE UPSTREAM THROUGH LINE L23



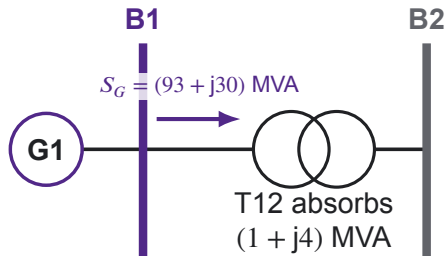
$$S_{23,S} = S_{23,R} + S_{L,\text{loss}} \\ = [(90 + j18) + (2 + j8)] \text{ MVA}$$

$$S_{23,S} = (92 + j26) \text{ MVA}$$

Sending-end and receiving-end flow differ whenever the branch model absorbs power.

Generator output

COMPLETE THE TRACE THROUGH TRANSFORMER T12



$$S_G = [(92 + j26) + (1 + j4)] \text{ MVA}$$

$$S_G = (93 + j30) \text{ MVA}$$

Generator output is a result of the complete downstream requirement in this trace.

System reconciliation

AUDIT THE COMPLETE TRACE

Real-power balance

$$P_G = P_D + P_{L,\text{loss}} + P_{T,\text{loss}}$$

$$93 = 90 + 2 + 1 \text{ MW}$$

Reactive-power balance

$$Q_G + Q_C = Q_D + Q_{L,\text{loss}} + Q_{T,\text{loss}}$$

$$30 + 12 = 30 + 8 + 4 \text{ Mvar}$$

A converged-looking result is not enough; supply, demand, support, and modeled absorption must reconcile.

System-data record

AN AUDITABLE RECORD COMPLETES THE ONE-LINE MODEL

Object	Supplied or selected	Model representation	Solved or checked
Generator	terminal bus, voltage target	capability and limits	P_G , Q_G , limit status
Bus	ID, nominal kV, attached objects	one nodal voltage state	$ V $, angle, balance
Transformer	buses, ratio, rating, tap	series impedance and tap	end flows, loss, loading
Line	endpoints, circuit ID, rating	R , X , charging B	end flows, loss, loading
Load	bus, P_D , Q_D , status	chosen demand model	served demand and current
Shunt	bus, Mvar rating, status	admittance or control	Q injection and voltage effect

A one-line becomes an analysis model only when each object is paired with supplied data, its model representation, and solved or checked quantities.